

HAMSHIKA RADHAKRISHNAN

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Education

University of Maryland, College Park

Master of Science, Data Science

Aug 2025 – May 2027

GPA: 4.0

Vellore Institute of Technology, Chennai, India

Bachelor of Technology, Computer Science with Specialization in AI and ML

Aug 2021 – Jun 2025

GPA: 3.8

Experience

OPENLANE

Data Scientist

Jun 2026 – Aug 2026

Indianapolis, Indiana

- Built and deployed a B2B vehicle recommender in **Snowflake** using **feature engineering, Cortex embeddings, and similarity scoring**; improved precision by **20%** and reduced runtime by **67%**.
- Productionized the recommendation pipeline via **dbt/GitHub** with daily automated runs and cold-start matching using dealer look-alikes by postal code and credit tier, generating **live bids and purchases**.
- Developed a **Logistic Regression** credit-risk model and scorecard for credit-line approvals, evaluating against the incumbent using **time-based splits, Precision, and Recall**.

Jocata Financial Advisory Technology

Data Scientist

Sep 2022 – Dec 2022

Hyderabad, India

- Designed and trained an **SSD-ResNet** model in **TensorFlow and PyTorch**, owning preprocessing, feature engineering, loss selection, and hyperparameter tuning to reach **92% accuracy** on structured document extraction and automated customer application filings.
- Optimized training and inference using **CUDA and GPU parallelism** on **NVIDIA hardware**, reducing latency by **30%**.
- Built **ETL and preprocessing pipelines** for large-scale noisy inputs and communicated model performance and extraction results to business and engineering stakeholders through **Tableau visualizations and reports**.

Research Experience

University of Maryland, College of Agriculture and Natural Resources

Research Assistant

May 2026 – Present

College Park, Maryland

- Built a **Monte Carlo Dropout BiLSTM** in **TensorFlow** to downscale GRACE water-storage anomalies from 1° to 0.25° across Andhra Pradesh and the Krishna Basin, achieving **$R = 0.93\text{--}0.94$ correlation** with GRACE observations.
- Implemented the research pipeline in **Python**, processing **2002–2022 hydro-climatic data** and generating **175K+ grid-month predictions** across **710 high-resolution cells**; reconstructed **1,451 grid-months** of missing GRACE observations.
- Performed **groundwater, drought, spatial, and uncertainty analyses**, achieving **83% SPEI-12 agreement**, delivered figures and analysis for manuscript review.

Projects

ALTA: Adaptive Logistics & Tracking Agent 🤖

Aug 2026

- Built an **8-node LangGraph agentic AI pipeline** on **GPT-4o** with conditional routing, dynamic **0–100 risk scoring**, and human-in-the-loop approval for normal, emergency, and rerouting shipment states across **900+ cargo routes**.
- Developed a real-time **FastAPI + React** dashboard with **React-Leaflet** and **Recharts**, deployed on **Render/Netlify**, with autonomous 3-stage rerouting system.

AI Music Generation: Comparative Study of Deep Learning Architectures 🎵

Dec 2024

- Implemented and trained **GAN, GRU, VAE, Transformer, and OpenAI Jukebox** models on sequential music data, evaluating **harmonicity, novelty, and coherence** across architectures.
- Processed **100+ MIDI files** using **tokenization, sequence padding, and normalization**; trained a **256-unit GRU** achieving **92% note prediction accuracy** and quantified the impact of dataset size, parameter count, and compute on model quality.

Echo of Hands - Gesture Recognition System 🖐️

May 2024

- Implemented a real-time sign-to-speech communication aid recognizing one-handed signs from webcam input using **TensorFlow, MediaPipe, and OpenCV**; trained a lightweight **5.4K-parameter MLP** on 2,470 samples with wrist-normalized hand-landmark features, achieving **97.9% accuracy**.
- Deployed the model fully **client-side in JavaScript**, reimplementing the TensorFlow forward pass with exported weights and matching outputs across all 2,470 samples with a maximum probability difference of 1.4×10^{-6} ; enabled **real-time, serverless inference**.

Technical Skills

Programming: Python, SQL, R, C++

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, HuggingFace Transformers, CUDA

Generative & Agentic AI: LangGraph, LangChain, OpenAI GPT-4o, RAG; architectures: Transformers, GANs, VAEs, LSTMs/GRUs

Data Engineering: Snowflake (Snowpark), dbt, PostgreSQL, MySQL, SQLite, Pandas, NumPy

Deployment & Visualization: FastAPI, Flask, React, Git, Render, Netlify, Tableau, Plotly, Matplotlib